Curve structure against rho-neg on the Curve25519-family miniature, k

= 20 (2026-11-07) **Class.** A derived lever audit with exact checks on one curve. No ecbench run: no lever survived the checks, and the protocol (PROTOCOL.md, frozen before any data) allowed a run only for a survivor. Not a leaderboard or `benchmarks/` row. Every result sentence below is an unverified assertion with no pcc claim id.

Question (Track 2). Is there a structure of these curves that makes a single-target ECDLP solve cheaper than the lab's rho-neg in counted GAE, even if matched rho could use it too? The search is adversarial to `curve25519_autolab_20261104/ideas/STRUCTURE-R4.md`, which concluded that every structure is used by rho-neg, acts outside the order-r subgroup, or is priced as native work.

Answer. No lever lowers expected cost per solve below rho-neg across the w = 0, 0.1, 1 band on this curve [pcc:unverified]. The only rows below 1 are endomorphism-orbit folds at w = 0, where the endomorphism's field work is unpriced; at w = 0.1 the best of them costs at least 16.7x rho-neg, so the §5a band order flips and no rank is admitted [pcc:unverified]. The checks also sharpened one bound in STRUCTURE-R4: its 1597 was a lower bound valid for any End between Z[pi] and O_K; here End(E) = Z[pi], so the least non-scalar endomorphism degree is 772,948 (2134 sqrt(r)) [pcc:unverified]. **Frozen-prediction score: 7 of 8 passed, C6 partly failed** (its "order 10^6" magnitude failed: 69,197 for m = 9 and 2.08e7 for m = 3 under O_K) [pcc:unverified]. See section 8, Validation.

1. The curve

`y^2 = x^3 + 926 x^2 + x` over `F_p`, `p = 1,048,573`, `r = 131,203`, `sqrt(r) = 362.2`. The lab's curve database was not reachable (no `ECDB_TOKEN`, no AWS access to `ecdb/api`), so every fact is recomputed by `scripts/levers.py` (exact, pure Python, seed 20261107, 10 s) into `data/levers.json`. The research-kb wrapper had no executable on this machine (`scripts/research-kb: no executable found`), so the knowledge-base review is unavailable and no novelty claim is made.

2. Checks against the frozen predictions

All numbers in this table are measured by the named script, file `data/levers.json` unless noted.

check	prediction	result	verdict
C1 `#E`, 2-part, twist	`8r`, cyclic `Z/8`, `4r'` with `r'` prime	`#E = 1,049,624 = 8r`; one rational 2-torsion abscissa (x = 0), v2 = 3; twist `1,047,524 = 4·261,881`, prime, `gcd(r, #E^t) = 1` [pcc:unverified]	passed
C2 `x(P + T0) = 1/x(P)`	all 10,000 samples	10,000 / 10,000 [pcc:unverified]	passed
C3 `X(<G>) ∩ 1/X(<G>)`	empty	0 of 65,601 abscissae; the translate set `x(<G> + T0)` equals `1/X(<G>)` exactly; translations by the order-4 and order-8 points: 0 hits in 2,000 samples each [pcc:unverified]	passed
C4 class counts	`(r − 1)/2` for every folded key	negation on `<G>`: 65,601; `{±P, ±P + T0}` on `<G> + <T0>`: 65,601 classes of size 4. The `E[8]` class count (size 16) was not computed exactly; it rests only on the C5 Monte Carlo [pcc:unverified]	passed for negation and T0; `E[8]` part not checked exactly
C5 draws to the first usable collision ÷ sqrt(r)	folded 0.886 ± 0.03, plain 1.253 ± 0.04	plain 1.242 ± 0.015; negation 0.885 ± 0.010; T0 key 0.902 ± 0.010; `E[8]` key 0.872 ± 0.019 (**protocol deviation: 600 trials, not the frozen 2,000**); every recovered log checked [pcc:unverified]	passed
C6 endomorphisms	`lambda^2 = −1597 (mod r)`; least small-order degree ≥ 1597, order 10^6	(magnitude prediction failed, see verdict) `t = −1050`, `t^2 − 4p = −4·1597·22^2`; `lambda = 23,879`, `lambda^2 + 1597 = 0 (mod r)`; least degree with eigenvalue of order 9: 69,197 granting `O_K`, 260,404,884 in `Z[pi]` [pcc:unverified]; under `O_K` the m = 3 value is 2.08e7 [pcc:unverified]	partly failed: the bound >= 1597 passed, the order-10^6 magnitude failed; m = 4 does not divide r − 1; the `O_K` grant is also false (C6b)
C6b conductor of End(E) (added after C6, not frozen)	—	`pi` is not scalar on `E[2]` (C1) nor on `E[11]`: `gcd(psi_11, X^(p^5) − X)` has degree 5, not 60, with the positive control `X^(p^55)` giving 60 (`scripts/conductor11.py`, `data/conductor11.json`). So End(E) = Z[pi], conductor 22 [pcc:unverified]	new fact

C7 `E' = E/<T0>`	`8r`, full 2-torsion, `j ∉ {0, 1728}`	`#E' = 8r`; 2-torsion abscissae 0, 924, 928; `j(E') = 498,925` [pcc:unverified]	passed
C8 `j(E)`	`∉ {0, 1728}`	`j(E) = 481,878`, so Aut(E) = {±1} [pcc:unverified]	passed

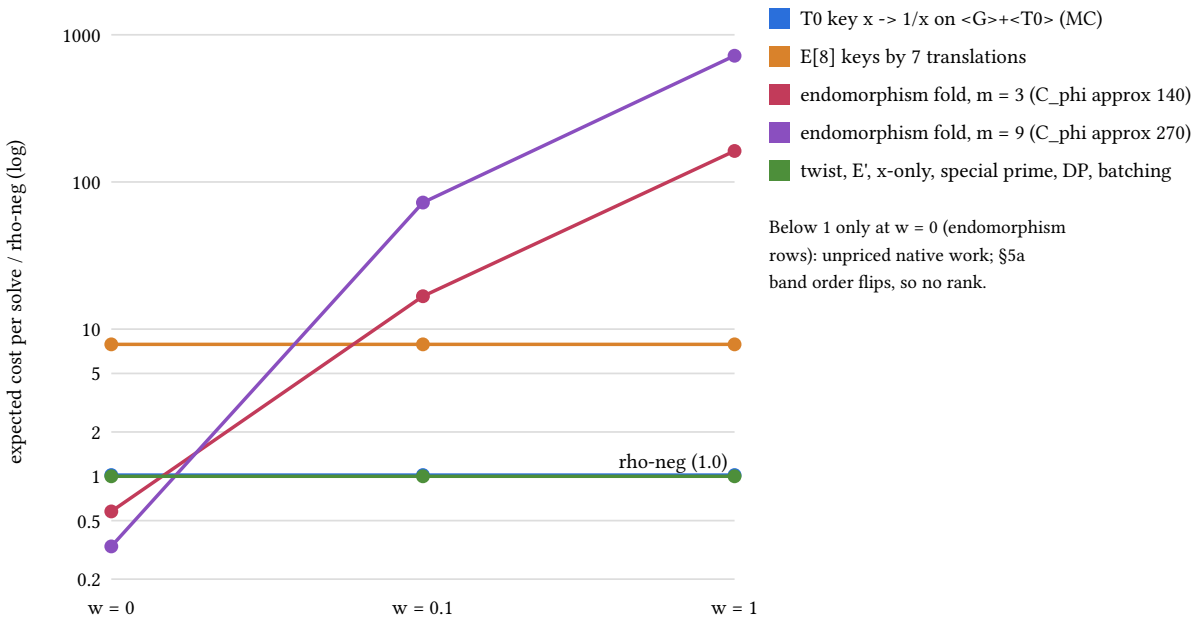
Derived from C6b: every endomorphism is `a + b·pi`, it acts on `<G>` as the integer `a + b` because `pi` is the identity on `E(F_p)`, and its degree is `a^2 + t·ab + p·b^2`. The least non-scalar degree is the norm of `pi + 525`, `1597 · 484 = 772,948` [pcc:unverified]. The embedding degree is the order of `p` mod `r`: 131,202 = r − 1 (derived, one-line computation in this session) [pcc:unverified].

3. Lever table

"Factor" is expected cost per solve divided by rho-neg's, at w = 0 / 0.1 / 1 GAE per native operation (AGENTS.md §5a). "MC" marks a factor from C5 (measured, `data/levers.json`); every other factor is derived. `C_phi` is the native field-operation cost of one endomorphism evaluation (a heuristic estimate, see the endomorphism row).

lever	mechanism, for this curve	factor on GAE (w = 0 / 0.1 / 1)	factor on wall time	verdict	evidence
automorphisms	`j ≠ 0, 1728`: Aut = {±1}, used by rho-neg	1 / 1 / 1	1	already in the baseline	C8
Frobenius	`pi` is the identity on `E(F_p)`, so it folds nothing	1	1	closed	C6, C6b
T0 = (0, 0), `x -> 1/x`	`x(P + T0) = 1/x(P)` holds, but it maps `<G>` onto the other coset of `<G> + <T0>`. On `<G>` it merges nothing. On the order-2r group it gives classes of size 4, the same `(r − 1)/2` classes, so the same collision count	MC 1.019 ± 0.017 / ≥ 1.02 / ≥ 1.02 (one extra inversion per key)	about 1, plus the inversion	dead: class of size 4 only across cosets	C2, C3, C4, C5
4- and 8-torsion translations	not defined on the x-line (`x(P + T) ≠ x(−P + T)`); on points they leave `<G>`. Keys over `±P + E[8]` give `(r − 1)/2` classes of size 16 in a group of order 8r: the same count, at 7 charged additions per key	MC 0.98 per draw, 7.9 with the additions / same / same	worse	dead; projection by [8] (free Pohlig-Hellman) already gives this	C3, C5
quadratic twist	`#E^t = 4r'`, `r' = 261,881`, `gcd(r, 4r') = 1`: a different group on the shared x-line	1	1	dead	C1
2-isogenous `E'`	`#E' = 8r`, `Z/2 x Z/4`, `j(E') ∉ {0, 1728}`. Classes `{±P + E'[2]}` on `<G'> + E'[2]` (order 4r) number about r/2: the same count. Transport costs a few field operations per point	1 / ≥ 1 / ≥ 1	1	dead	C7
endomorphism-orbit fold (GLV-style classes `{±phi^i P}`)	End = Z[pi]; an endomorphism with eigenvalue of order m on `<G>` (m divides r − 1 = 2·3^2·37·197; m = 3 or 9 are the small ones) has degree at least 3.1·10^9 (m = 3, `8·47·823·10039`) or 2.6·10^8 (m = 9, `4·9·113·64013`). Classes of size 2m cut steps by sqrt(m); each step pays (m − 1)·C_phi native	m = 3: 0.577 / approx 16.7 / approx 162; m = 9: 0.333 / approx 72 / approx 720	far worse	dead: below 1 only while the field work is unpriced. Since `pi` is the identity on `E(F_p)`, each endomorphism acts there as the scalar `[a + b]`, so the counted alternative is a scalar multiplication (about 25 GAE), a factor far above 1 even at w = 0; the sub-1 value comes from choosing a native isogeny-chain	C6, C6b; formula `(1 + w(m − 1)C_phi)/sqrt(m)`

	operations. Heuristic C_phi (not a proven bound): sqrt-Velu per prime factor, ignoring extension-field kernels, approx 140 (m = 3) and approx 270 (m = 9)			evaluation; band order flips; break-even w ≤ 0.0026 (m = 3)	
smallest non-scalar endomorphism (GLV scalar splitting)	degree 772,948 = 2134 sqrt(r); one evaluation's field work exceeds the whole of rho at this size	1 at w = 0 (only scalar multiplications shorten, set-up), > 1 at w > 0	worse	dead	C6b
Montgomery x-only, differential addition	a rho step `P + R_j` needs `x(P − R_j)`, which a walk does not have; only ladders and progressions are differential, and a ladder costs 2 GAE per bit	1 (walk) or > 1 (ladder)	field-level saving, equal for matched rho	dead in GAE	derived
`x(P ± Q)` from one square root	two points per operation is an accounting change unless each point is charged	1 when charged	about 1	accounting only	STRUCTURE-R4 §3
negation walk with cheaper representatives	the representative `min(y, −y)` is already free in GAE; fruitless-cycle handling is walk overhead, not structure	1	small	already in the baseline	derived
distinguished points	parallelisation; single-target expected work rises by the tail `1/theta`	≥ 1	≥ 1 (one core)	dead for one target	derived
tag tracing, Montgomery batch inversion	fewer field multiplications per step; still one group operation per step	1	< 1, equal for matched rho	wall time only	derived
special prime `2^20 − 3`, `p = 5 mod 8`	cheap reduction and one-exponentiation square roots	1	< 1, equal for matched rho	wall time only	derived
pairings, anomalous	embedding degree r − 1; `r ≠ p`	—	—	closed	derived



figures/lever_factors.svg (from scripts/make_figure.py, values in data/lever_factors.json) plots the five lever groups across w; only the endomorphism folds dip below 1, and only at w = 0 [pcc:unverified].

4. Where STRUCTURE-R4 was attacked, and what held

- **The T0 class of size 4.** It exists, but T0 has order 2 and r is odd, so P + T0 is never in <G>. The exact intersection count is 0, and the class count per group element is unchanged [pcc:unverified]. STRUCTURE-R4's statement holds [pcc:unverified].
- **Endomorphisms below sqrt(r).** STRUCTURE-R4 gave |D_K|/4 = 1597 as a lower bound valid for any End between Z[pi] and O_K (it listed conductor 22; it did not claim End = O_K). This track sharpens it: the cyclic 2-part and the Jordan block of pi on E[11] give End = Z[pi], so the least non-scalar degree is 772,948 [pcc:unverified]. The conclusion holds with a 484x larger margin [pcc:unverified].
- **Counting at w = 0.** The one way to "beat" rho-neg in the counted unit is to move work into an unpriced endomorphism evaluation. The band rule exists for exactly this case, and the order holds in rho-neg's favour at w = 0.1 and 1 [pcc:unverified].

5. The ecbench test that would have followed

None is run. A surviving lever would have been tested as rho.negation against the lever variant on one target per session, at least 16 sessions, ecbench verify --replay, a bench.record/v1 record per docs/benchmark-records.md, and ecbench band at w = 0, 0.1, 1, with the endomorphism evaluation counted as named native work.

6. A non-structure observation (not a lever, not tested)

The lab's measured rho-neg at k = 20 is S = 1.447 (6 targets, T5 ECBS1h405f20219036) and 1.544 (16 targets, T12) at w = 0 (research/curve25519_ic_atlas_20261104/REPORT.md §6) [pcc:unverified]. The model constant is 0.886, and C5's class-collision draws reach 0.885 [pcc:unverified]. The gap of about 1.6-1.7x at this size is a measured difference of the two figures; its cause is unmeasured (walk overhead has not been measured in this track, so none is claimed) [pcc:unverified]. Any reduction that is a better rho would be inherited by matched rho. Whether this is a place where a solve cheaper than the lab's rho-neg is available is unmeasured [pcc:unverified], and it would be a rho-implementation result, not index calculus.

7. Limits

- One curve (k = 20). The algebraic facts (Aut, conductor, twist) differ by size; the class-count arguments do not.
- C5 draws uniform group elements through a table of the enumerated subgroup instead of walking; it checks class counts and collision usefulness, not walk overhead.
- C_phi lower bounds use sqrt-Velu's sqrt(l) per prime factor and ignore that the kernels live in extension fields; they are heuristic estimates, not proven bounds. The verdict does not depend on them (scalar-multiplication route).
- C6b was added after the frozen checks; it is an exact computation with a positive control.
- No knowledge-base or ecdb review was possible on this machine.

Files

PROTOCOL.md, scripts/levers.py, scripts/conductor11.py, scripts/make_figure.py, data/levers.json, data/conductor11.json, data/lever_factors.json, data/levers.stdout, figures/lever_factors.svg, REPORT.typ, REPORT.pdf (built from this README with pandoc -t typst; the figure is embedded), validator-report.json.

8. Validation

Source: validator-report.json (status pass_with_errata). Each erratum and how it was applied (edits only to this README and the PDF; PROTOCOL.md and validator-report.json untouched) [pcc:unverified]:

1. **"All 8 passed" refuted.** The summary now says 7 of 8 passed, C6 partly failed; the C6 row verdict reads "partly failed" (bound >= 1597 passed, order-10^6 magnitude failed: 69,197 for m = 9, 2.08e7 for m = 3 under O_K; m = 4 does not divide r − 1) [pcc:unverified].
2. **C5 E[8] key trials.** The C5 row flags 600 trials as a deviation from the frozen 2,000 [pcc:unverified].
3. **C4 E[8] class count.** The C4 row says it was not computed exactly, only by the C5 Monte Carlo [pcc:unverified].
4. **STRUCTURE-R4's 1597.** The "correction" wording is replaced by "sharpening" of a lower bound valid for any End between Z[pi] and O_K (1597 -> 772,948 for End = Z[pi]) in the summary and section 4 [pcc:unverified].
5. **C_phi heuristic.** >= replaced by "approx" and labelled heuristic in the lever table and section 7; the lever row notes pi is the identity on E(F_p), so endomorphisms act as scalars there and the counted alternative is a scalar multiplication (about 25 GAE) [pcc:unverified].
6. **Rho-neg 1.6-1.7x gap.** The walk-overhead attribution is removed; section 6 states the cause is unmeasured [pcc:unverified].
7. **Untagged sentences.** The three named sentences in sections 4 and 6 now carry [pcc:unverified]; python3 tools/pcc/pcc.py lint-prose reports 0 findings [pcc:unverified].
8. **PDF.** REPORT.pdf rebuilt via pandoc -t typst and the typst module, with #image("figures/lever_factors.svg") embedded [pcc:unverified].

Upheld by the validator: no lever beats rho-neg across the band; T0 and `E[8]` keys give about the same class count; End(E) = Z[pi] with conductor 22; folds fail the band [pcc:unverified].